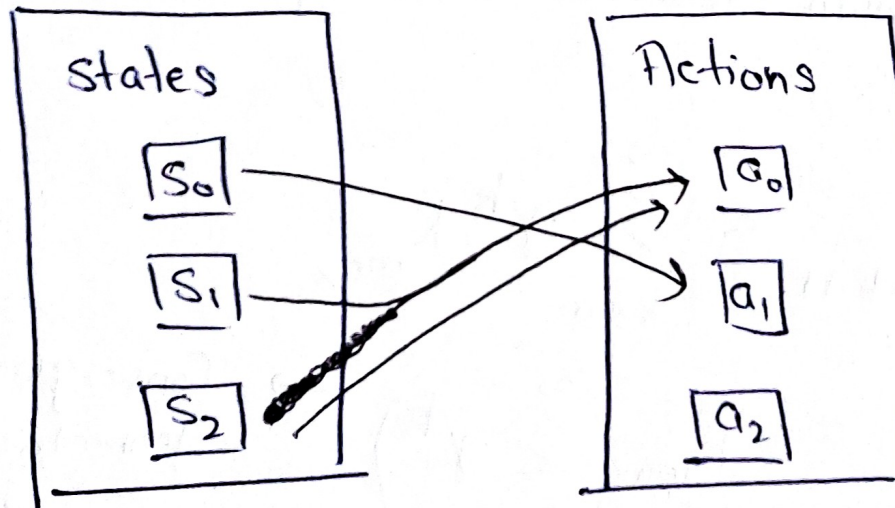


Week 3 →

Specifying policies → Policy is a distribution over actions for each possible state.

1) Deterministic policy

$$\pi(s) = a$$



In other words, policy maps the current state onto a set of probabilities for taking each action.

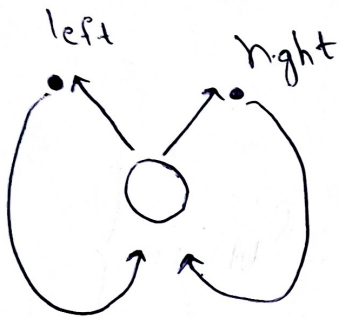
2) Stochastic policy -

(7)

$$\sum_{a \in \mathcal{A}(s)} \pi(a|s) = 1$$

$$\pi(a|s) \geq 0$$

Valid and Invalid policies →



1) left with 50% probability
and right with 50% probability

Valid policy since it depends
on current state only!

(Valid policy should only depend only
on current state not on other
state (like time or previous state).

2) Alternate policy → Not valid policy!

Since it is conditional on
the last action.

Value functions →

⇒ In Reinforcement learning, reward captures the notation of short term. The objective however, is to learn a policy that achieves the most reward in the long run.

Value functions formalize what this means!

State value functions →

$$V(s) \doteq E[G_t | S_t = s] \quad \left\{ G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \right\}$$

⇒ It is the future reward an agent can expect to receive starting from a particular state.

i.e. It is the expected return from a given state!

$$V_{\pi}(s) \doteq E_{\pi}[G_t | S_t = s]$$

Value function is defined with respect to the given policy

It indicates that the value function is contingent on the agent selecting actions according to π .

Action Value function →

(8)

$$q_{\pi}(s, a) \doteq E_{\pi} [G_t | S_t = s, A_t = a]$$

✗ An action value describes what happens when the agent selects a particular action.

⇒ The action value of a state is the expected return if the agent selection action 'a' and then follows policy π .

✗ The value functions summarizes all the possible futures by averaging over returns.

✗ Value functions predict rewards into the future.

✗ Value functions enable us to judge the quality of different policies.

✗ The state value is equal to the expected sum of future rewards.

Bellman Equations →

1) State Value Bellman equation:

$$V_{\pi}(s) \doteq E_{\pi} [G_t | S_t = s]$$

$$\left\{ G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \right\}$$

⇒ It defines the relationship b/w the value of a state and the value of its possible successor states.

Since, $G_t = R_{t+1} + \gamma G_{t+1}$

$$V_{\pi}(s) = E_{\pi} [R_{t+1} + \gamma G_{t+1} | s_t = s]$$

$$V_{\pi}(s) = \sum_a \pi(a|s) \left(\sum_{s'} \sum_a p(s', a | s, a) \left\{ \begin{aligned} & \gamma + \gamma E_{\pi} [G_{t+1} | s_{t+1} = s'] \end{aligned} \right\} \right)$$

$$V_{\pi}(s) = \sum_a \pi(a|s) \left\{ \sum_{s'} \sum_a p(s', a | s, a) (\gamma + \gamma V_{\pi}(s')) \right\}$$

2) Action-Value Bellman equation $\rightarrow$

$$Q_{\pi}(s, a) = E_{\pi} [G_t | s_t = s, A_t = a]$$

$$Q_{\pi}(s, a) = \sum_{s'} \sum_a p(s', a | s, a) \left\{ \gamma + \gamma E_{\pi} [G_{t+1} | s_{t+1} = s'] \right\}$$

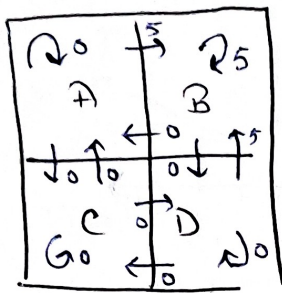
$$Q_{\pi}(s, a) = \sum_{s'} \sum_a p(s', a | s, a) \left\{ \gamma + \gamma \right.$$

$$\left. \sum_{a'} \pi(a'|s') E_{\pi} [G_{t+1} | s_{t+1} = s', A_{t+1} = a'] \right\}$$

$$Q_{\pi}(s, a) = \sum_{s'} \sum_a p(s', a | s, a) \left\{ \gamma + \gamma \sum_{a'} \pi(a'|s') \cdot Q_{\pi}(s', a') \right\}$$

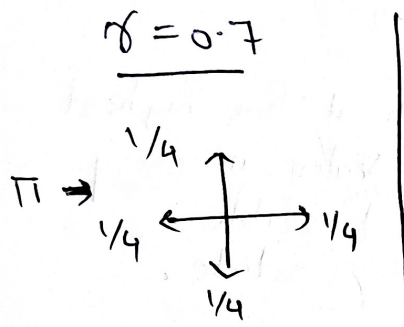
Why Bellman Equations?

(9)



$$V_{\pi}(s) = \sum_a \pi(a|s) \sum_{s'} p(s', a | s, a) \cdot [r + \gamma V_{\pi}(s')]$$

→ Given s and a → s' and r both are deterministic in this case.



$$V_{\pi}(s) = \sum_a \pi(a|A) [r + 0.7 V_{\pi}(s')]$$

$$V_{\pi}(A) = \frac{1}{4} (5 + 0.7 V_{\pi}(B)) + \frac{1}{4} 0.7 V_{\pi}(C) + \frac{1}{4} 0.7 V_{\pi}(A)$$

... similar eqn can be written for all four states

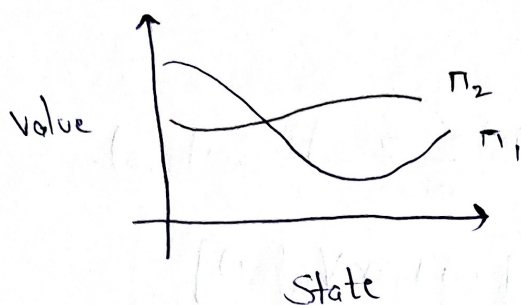
Solving all four linear algebraic eqn we have -

$$\left. \begin{aligned} V_{\pi}(A) &= 4.2 \\ V_{\pi}(B) &= 6.1 \\ V_{\pi}(C) &= 2.2 \\ V_{\pi}(D) &= 4.2 \end{aligned} \right\}$$

unmanageable

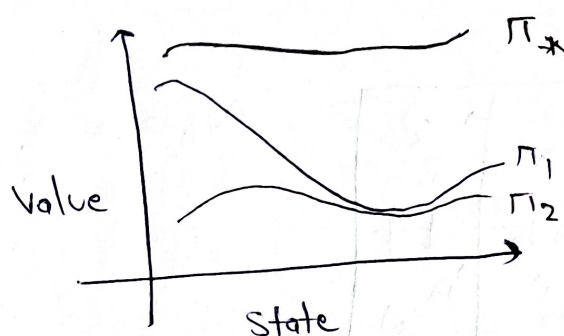
⇒ Thus Bellman equations turns infinite sum over possible futures, to a simple linear algebra problem.

Optimal Policies →



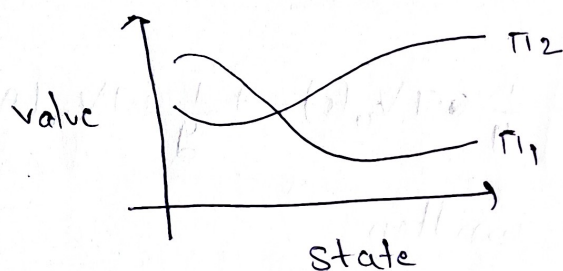
$$\pi_1 \neq \pi_2$$

$$\pi_2 \neq \pi_1$$



$$\pi_1 \geq \pi_2$$

✗ An optimal policy π_* is as good as or better than all other policies. } i.e. it has highest value in every possible state.

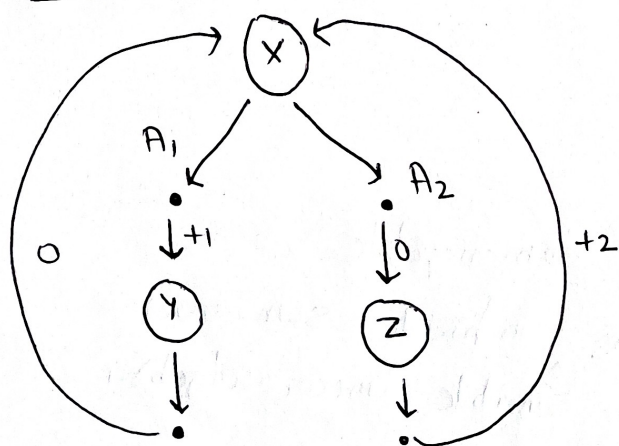


$$\pi_3(s) = \pi_1(s)$$

$$\pi_3(s) = \pi_2(s)$$

therefore we will always have the optimal policy!

example



$$\pi_1(x) = A_1$$

$$\pi_2(x) = A_2$$

$$1) \gamma = 0$$

therefore

$$V_{\pi_1}(x) = 1$$

$$V_{\pi_2}(x) = 0$$

Hence π_1 is the optimal policy in this case. 1

$$2) \gamma = 0.9$$

(10)

$$V_{\pi_1}(x) = 1 + 0.9 V_{\pi_1}(y)$$

$$V_{\pi_1}(x) = 1 + 0.9 [0 + 0.9 V_{\pi_1}(x)]$$

$$V_{\pi_1}(x) = 1 + (0.9)^2 V_{\pi_1}(x)$$

$$V_{\pi_1}(x) = 1 + (0.9)^2 [1 + (0.9)^2 V_{\pi_1}(x)]$$

$$\Rightarrow V_{\pi_1}(x) = \sum_{k=0}^{\infty} (0.9)^{2k}$$

$$V_{\pi_1}(x) = \frac{1}{1 - (0.9)^2} \approx 5.3$$

Similarly for $V_{\pi_2}(x) = 0 + 0.9 V_{\pi_2}(z)$

$$V_{\pi_2}(x) = 0.9 (2 + 0.9 V_{\pi_2}(x))$$

$$V_{\pi_2}(x) = \frac{2 \times 0.9}{1 - (0.9)^2} \approx 9.5$$

hence π_2
is the
optimal
policy
here!

There may be more than
one optimal policies possible

Why?

(for example, if for a given state, multiple actions
have equal probabilities to get chosen.)

Optimal Value functions $\rightarrow$

Recall that,

$\pi_1 \geq \pi_2$ if and only if $V_{\pi_1}(s) \geq V_{\pi_2}(s)$ for all $s \in \mathcal{S}$

$$V_{\pi^*}(s) = E_{\pi^*} [G_t | s_t = s]$$

$$V_{\pi^*}(s) = \max_{\pi} V_{\pi}(s) \quad \forall s \in \mathcal{S}$$

$\Rightarrow$ All optimal policies have same optimal state value function. — denoted by (V^*)

$\Rightarrow$ All optimal policies also share the same optimal action value function — denoted by (q^*)

$$q_{\pi^*}(s, a) = \max_{\pi} q_{\pi}(s, a) \quad \forall s \in \mathcal{S} \text{ and } a \in \mathcal{A}$$

$$\Rightarrow V_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', a'} p(s', a' | s, a) [r + \gamma V_{\pi}(s')]$$

$$V_*(s) = \sum_a (\pi_*(a|s)) \sum_{s', a'} p(s', a' | s, a) [r + \gamma V_*(s')]$$

$$V_*(s) = \max_a \sum_{s', a'} p(s', a' | s, a) [r + \gamma V_*(s')]$$

(optimal policy will assign probability 1, for an action that achieves the highest value and probability 0, for all other actions.)

=> above eqn is called Bellman optimality equation for v^* . (11)

=> Doing similar things for action value functions —

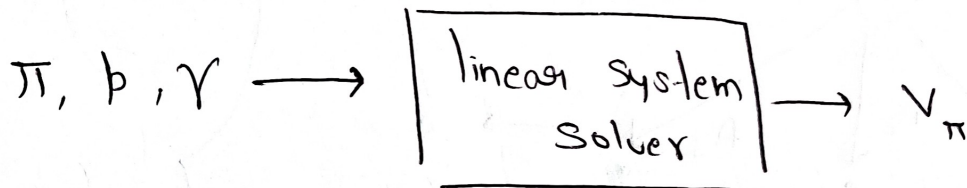
$$q_{\pi}(s, a) = \sum_{s'} \sum_{a'} p(s', a' | s, a) \left[r + \gamma \sum_{a''} \pi(a'' | s') q_{\pi}(s', a'') \right]$$

$$q_*(s, a) = \sum_{s'} \sum_{a'} p(s', a' | s, a) \left[r + \gamma \sum_{a''} \pi^*(a'' | s') q_*(s', a'') \right]$$

$$\Rightarrow q_*(s, a) = \sum_{s'} \sum_{a'} p(s', a' | s, a) \left[r + \gamma \max_{a''} q_*(s', a'') \right]$$

Bellman optimality eqn for q^* .

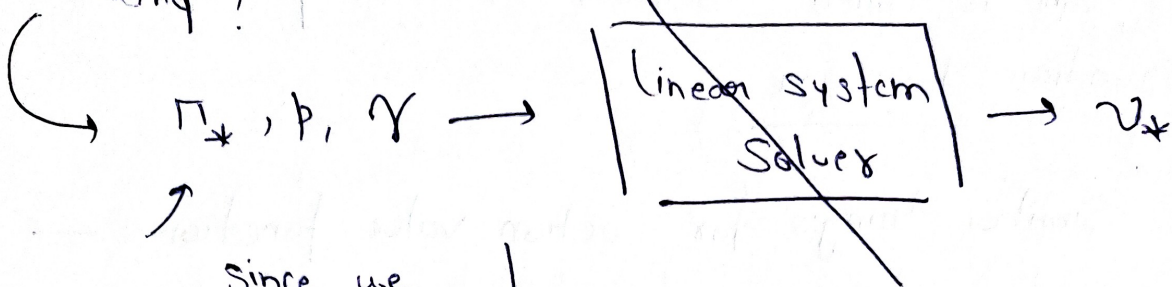
Note: previously we ~~solved the Bellman equations~~ compute the value functions and policies using γ linear equations. (ie by solving system of bellman equations)



However,

bellman optimality equation cannot be solve the way as 'max' is not linear function!

why?



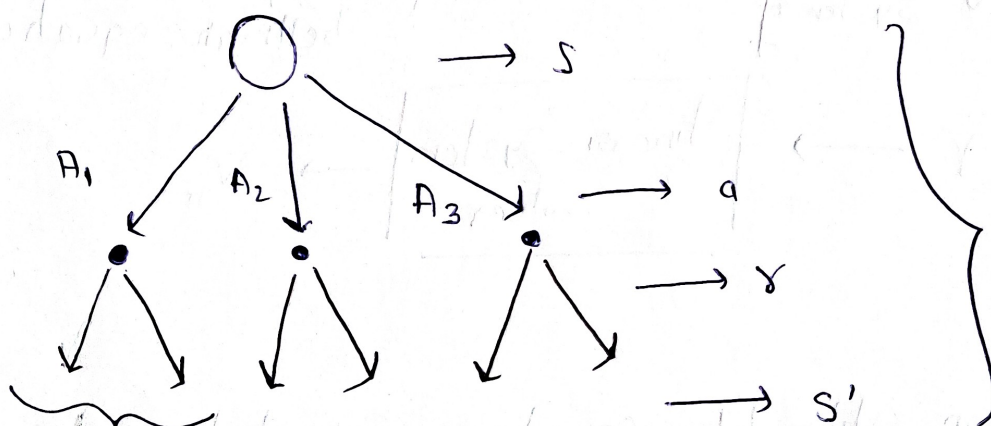
Since we
don't know
 π_*

if we know π_* , then we
know the fundamental goal
of ~~the~~ Reinforcement learning.

$\Rightarrow$ Using optimal ~~polices~~ Value functions to get
Optimal policies $\rightarrow$

$$V_*(s) = \max_a \sum_{s'} \sum_{\pi} p(s', \pi | s, a) [\pi + \gamma V_*(s')]$$

$$\pi_*(s) = \arg \max_a \sum_{s'} \sum_{\pi} p(s', \pi | s, a) [\pi + \gamma V_*(s')]$$



two possible
next state
after taking
action A_1

$$\Rightarrow \left\{ \pi_*(s) = \max_a q_*(s, a) \right\} \quad (12)$$

Summary :

⇒ Bellman equations can be directly solved to find the value function. These bellman equations help us evaluate the policies. however they do not yet achieve our ultimate goal.

i.e to find the ~~perfect~~ policy that attains as much reward as possible.

⇒ Therefore we define optimal policies, the optimal value function and the associated Bellman's optimality equations